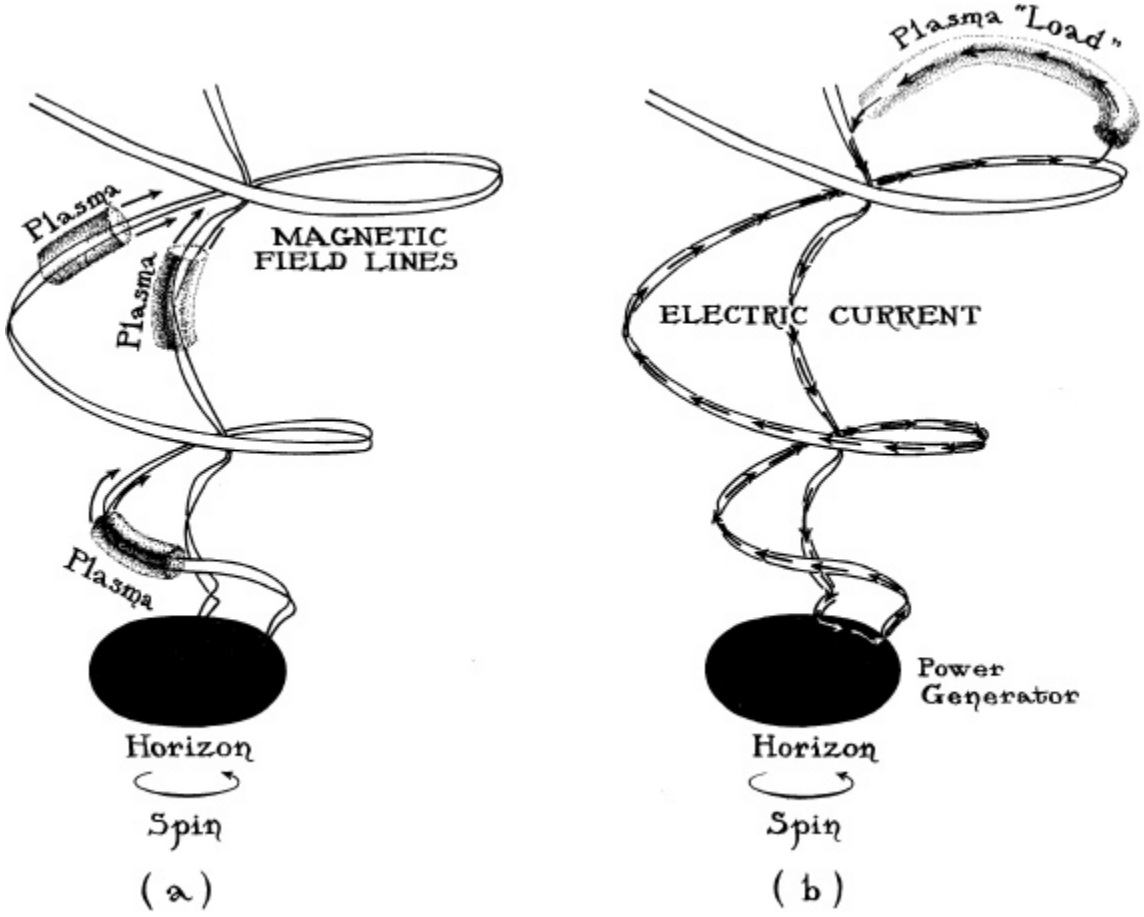
From this viewpoint (Figure 11.4b), it is the power carried by the circuit that accelerates the plasma to form jets. From the viewpoint of Chapter 9 (Figure 11.4a), it is the spinning magnetic field lines, whipping around and around, that accelerate the plasma. The two viewpoints are just different ways of looking at the same thing. The power comes ultimately from the hole's spin in both cases. Whether one thinks of the power as carried by the circuit or as carried by the spinning field lines is a matter of taste.



11.4 Two viewpoints on the *Blandford–Znajek process* by which a spinning, magnetized black hole can produce jets. (a) The hole's spin creates a swirl of space which forces magnetic fields threading the hole to spin. The spinning fields' centrifugal forces then accelerate plasma to high speeds (compare with Figure 9.7d). (b) The magnetic fields and the swirl of space together generate a large voltage difference between the hole's poles and equator; in effect, the hole becomes a voltage and power generator. This voltage drives current to flow in a circuit. The circuit carries electrical power from the black hole to the plasma, and that power accelerates the plasma to high speeds.

The electric circuit description, although based on the standard curved spacetime laws of physics, was totally unexpected, and the flow of current